

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

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PROGRAMMING LOGIC ASSIGNMENT

Designing a Logical System to Improve Access to Education in Sierra Leone Using Computational Thinking

Student Name: _____

Student ID: _____

Department: _____

Course: Programming Logic

Lecturer: _____

Institution: _____

Date: _____

Note: This assignment focuses on logical system design and does not require source code.

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1. Abstract

Education is an important foundation for national development because it provides people with the knowledge and skills needed to improve their lives and contribute to society. In Sierra Leone, some schools and communities face challenges such as inadequate learning materials, limited access to computers and the internet, overcrowded classrooms, insufficient school facilities, and difficulties monitoring students' academic progress. These challenges can affect the quality of education, particularly in rural and underserved communities.

As a junior system analyst, this assignment proposes an Education Access and Learning Support System (EALSS) designed to help schools identify educational needs, manage learning resources, monitor student attendance, and improve access to educational materials. The proposed system uses computational thinking, algorithm design, pseudocode, flowcharts, decision structures, functions, and modular design to explain how the solution would operate. It also examines how open-source software, Digital Public Goods (DPGs), and version-control tools such as Git and GitHub can support the development and management of the system.

The proposed solution supports Sustainable Development Goal 4 (Quality Education) and contributes to SDG 9 (Industry, Innovation and Infrastructure) and SDG 10 (Reduced Inequalities). It provides a practical logical design that could be adapted to the needs of schools and communities in Sierra Leone.

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2. Introduction

Education plays a major role in reducing poverty, creating employment opportunities, improving literacy, and promoting social and economic development. A country with an effective education system is better positioned to develop skilled workers, responsible citizens, and future leaders.

However, access to quality education can be difficult for some communities in Sierra Leone. Challenges may include shortages of textbooks, limited computer facilities, unreliable electricity, poor internet connectivity, inadequate classrooms, and difficulties keeping accurate student records. Students in remote communities may face additional barriers to accessing digital learning materials and educational support.

Traditional methods of recording attendance, distributing learning materials, and monitoring students can also create administrative difficulties when records are kept on paper or information is not shared effectively.

To address these challenges, the government and educational institutions can adopt a logical, affordable, and accessible digital solution. The proposed Education Access and Learning Support System would help schools record important information, identify students who need support, monitor available resources, and distribute suitable learning materials.

2.1 Aim of the Assignment

To design a logical and affordable system that improves access to educational resources and supports the monitoring of students' learning needs in Sierra Leone.

2.2 Objectives

- Analyse education challenges using computational thinking.
- Identify the inputs, processes, and outputs of the proposed system.
- Design an efficient algorithm using pseudocode.
- Develop flowcharts that explain the system's decision-making process.
- Apply decision structures, functions, and modular design.
- Explain the roles of open-source software, Digital Public Goods, Git, and GitHub.
- Demonstrate how the proposed system supports the Sustainable Development Goals.

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3. Conceptual Analysis

3.1 Analysis of the Education Problem

The first responsibility of a junior system analyst is to understand the problem before proposing a solution. The analyst must identify the people affected, the causes of the problem, the available resources, and the expected results.

- Limited learning materials: Some schools may not have enough textbooks, revision materials, or other learning resources.
- Limited access to technology: Some communities lack computers, reliable electricity, or affordable internet access.
- Poor record management: Paper-based attendance and resource records can be lost, damaged, or difficult to update.
- Difficulty identifying students who need help: Teachers may find it difficult to identify repeated absences or shortages of essential learning materials.
- Unequal educational opportunities: Students in remote communities may have fewer opportunities to access digital learning resources than students in better-connected areas.

These problems are connected. For example, a school with insufficient learning materials and poor record management may struggle to identify which classes need additional textbooks. A logical digital system can help organise this information and support better decisions.

3.2 Proposed Solution

The proposed Education Access and Learning Support System (EALSS) would be a simple information system that schools can use to record student attendance, track learning materials, identify shortages, and provide access to educational resources.

- Student attendance management.
- Learning-material inventory management.
- Identification of students requiring additional support.
- Distribution of digital and offline learning materials.
- Generation of reports for school administrators and education authorities.

The system should be designed for schools with different levels of technological access. Schools with reliable internet could use an online version, while schools with limited connectivity could use offline records and synchronise information when a connection

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becomes available.

The system would support educational decision-making rather than replace teachers or school administrators. Decisions involving individual students would remain subject to appropriate human review.

3.3 Application of Computational Thinking

Computational thinking involves breaking down a complex problem into smaller parts, identifying patterns, focusing on important information, and designing step-by-step solutions. The four major techniques are decomposition, pattern recognition, abstraction, and algorithm design.

A. Decomposition

Decomposition means dividing a large problem into smaller, manageable components. The education problem can be divided into student registration and attendance; management of textbooks and other learning materials; access to digital learning resources; identification of students needing assistance; and reporting and evaluation. Each component can be designed and tested separately before being combined into the complete system.

B. Pattern Recognition

Pattern recognition involves identifying similarities or repeated occurrences in information. A school may discover that certain classes frequently run short of textbooks, some students repeatedly miss school, schools in remote communities experience regular internet interruptions, or particular subjects require additional learning materials. Recognising these patterns helps administrators identify recurring problems and plan interventions. Repeated absence should trigger a supportive review rather than an automatic conclusion that a student is neglecting education.

C. Abstraction

Abstraction means concentrating on essential information while ignoring unnecessary details. Important information might include a student identification number, class and school, attendance records, available learning materials, learning-resource requirements, and support actions taken. The system does not need to collect unrelated personal information. Limiting data collection makes the system easier to manage and helps protect students' privacy.

D. Algorithm Design

Algorithm design involves creating a clear sequence of steps to solve a problem. For textbook management, the algorithm receives inventory information, identifies the available and required quantities, compares them, determines whether a shortage exists, records any shortage, generates a report, and allows an administrator to review the report and arrange an appropriate response.

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4. System Requirements and Logical Design

4.1 Functional Requirements

Requirement	Description
Student records	Store necessary student and class information.
Attendance tracking	Record daily attendance and identify repeated absences for review.
Resource management	Record textbooks and other learning materials available.
Shortage detection	Compare available materials with the required quantity.
Learning access	Provide suitable online and offline learning materials.
Report generation	Produce summaries of attendance and resource shortages.

4.2 Non-Functional Requirements

- Usability: Teachers and administrators should be able to use the system with basic training.
- Affordability: Use low-cost equipment and freely available software where appropriate.
- Reliability: Protect records against accidental loss.
- Accessibility: Support users with different abilities and levels of digital literacy.
- Security: Restrict access to student records to authorised users.
- Offline capability: Keep essential functions available where internet connectivity is unreliable.

4.3 Input, Process, and Output

The system follows the Input–Process–Output (IPO) model.

Stage	Examples
Input	Student attendance, class information, textbook stock, and resource requirements.
Process	Validate records, compare quantities, check attendance patterns, and apply decision rules.
Output	Attendance reports, shortage alerts, learning-resource access, and recommended support actions.

For example, if a school requires 100 mathematics textbooks but has only 65, the system calculates a shortage of 35 textbooks and includes the result in its report.

5. Algorithm Design Using Pseudocode

An algorithm is a finite sequence of logical steps used to solve a problem. Pseudocode describes those steps in a language-independent form without requiring a specific programming language.

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5.1 Algorithm for Identifying Textbook Shortages

Problem: A school needs to know whether it has enough textbooks. The inputs are the number available and the number required. The process compares the quantities, and the output reports whether a shortage exists and its size.

```
BEGIN
    INPUT textbooks_available
    INPUT textbooks_required

    IF textbooks_available < 0 OR textbooks_required < 0 THEN
        DISPLAY "Invalid quantity"
    ELSE
        IF textbooks_available < textbooks_required THEN
            shortage = textbooks_required - textbooks_available
            DISPLAY "Textbook shortage detected"
            DISPLAY shortage
        ELSE
            DISPLAY "Sufficient textbooks available"
        ENDIF
    ENDIF
END
```

This pseudocode uses a sequence of instructions and nested decision structures. It also checks that quantities are not negative. If 65 textbooks are available and 100 are required, the shortage is $100 - 65 = 35$ textbooks.

5.2 Algorithm for Identifying Students Who Need Attendance Support

Problem: A school needs to identify students whose repeated absences require attention. The algorithm calculates attendance percentage and compares it with an agreed threshold.

```
BEGIN
    INPUT days_attended
    INPUT total_school_days
    SET attendance_threshold = 80

    IF total_school_days <= 0 THEN
        DISPLAY "Invalid school-day total"
    ELSE IF days_attended < 0 OR
            days_attended > total_school_days THEN
        DISPLAY "Invalid attendance record"
    ELSE
        attendance_rate =
            (days_attended / total_school_days) * 100

        IF attendance_rate < attendance_threshold THEN
            DISPLAY "Attendance review required"
            CALL ReferForSupport()
        ELSE
            DISPLAY "Attendance meets the threshold"
        ENDIF
    ENDIF
END
```

The 80% threshold is an illustrative design choice, not a statement of Sierra Leone's official attendance policy. A school or education authority would need to approve an appropriate threshold. The ReferForSupport() function allows a teacher or authorised school officer to review the student's circumstances and consider support, such as

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contacting a guardian or identifying transport or learning difficulties.

5.3 Why These Algorithms Are Efficient

The textbook algorithm requires a small number of comparisons and arithmetic operations. Its processing time remains constant as the number of textbooks increases, provided the input is already available. The attendance algorithm also uses a fixed sequence of calculations and decisions for each student. For a large school, the procedure can be repeated for each student. Both algorithms include input validation to prevent invalid values from being processed as correct records.

6. Professional Flowcharts

A flowchart is a graphical representation of an algorithm. It uses standard symbols to show the order of operations and the direction of decisions. An oval represents start or end; a parallelogram represents input or output; a rectangle represents a process; a diamond represents a decision; and arrows show the direction of the process.

6.1 Flowchart for Textbook Shortage Detection

```
flowchart TD
    A([Start]) --> B[/Enter available and required textbooks/]
    B --> C{Are both quantities valid?}
    C -- No --> D[/Display invalid quantity/]
    D --> H([End])
    C -- Yes --> E{Available less than required?}
    E -- Yes --> F[Calculate shortage]
    F --> G[/Display shortage and generate report/]
    G --> H
    E -- No --> I[/Display sufficient stock/]
    I --> H
```

This Mermaid flowchart can be copied into https://mermaid.live/ or recreated in draw.io. The process begins by receiving available and required quantities. The first decision checks whether the input is valid. Invalid quantities produce an error message. Valid quantities are compared: a shortage produces a calculation and report, while sufficient stock produces a confirmation.

6.2 Flowchart for Attendance Monitoring

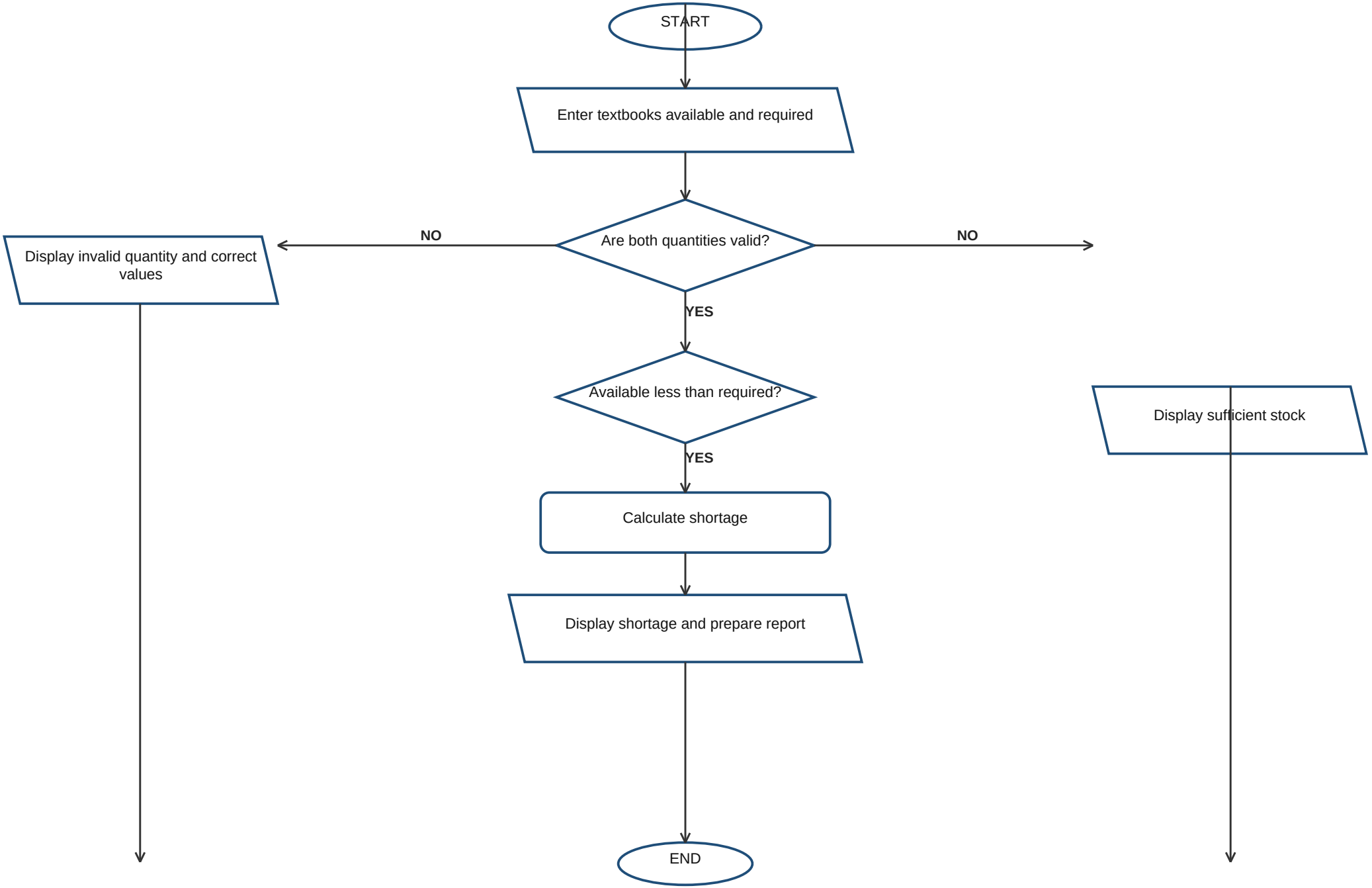
```
flowchart TD
    A([Start]) --> B[/Enter attendance information/]
    B --> C{Is the record valid?}
    C -- No --> D[/Request record correction/]
    D --> B
    C -- Yes --> E[Calculate attendance percentage]
    E --> F{Below approved threshold?}
    F -- Yes --> G[Refer for supportive review]
    G --> H[/Record review outcome/]
    F -- No --> I[/Record satisfactory attendance/]
    H --> J([End])
    I --> J
```

This flowchart demonstrates input validation, calculation, decision-making, and error correction. Invalid records return to the input stage for correction. Valid records are evaluated against an approved attendance threshold. Students requiring attention are referred for supportive review rather than being automatically punished.

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6.1 Flowchart for Textbook Shortage Detection

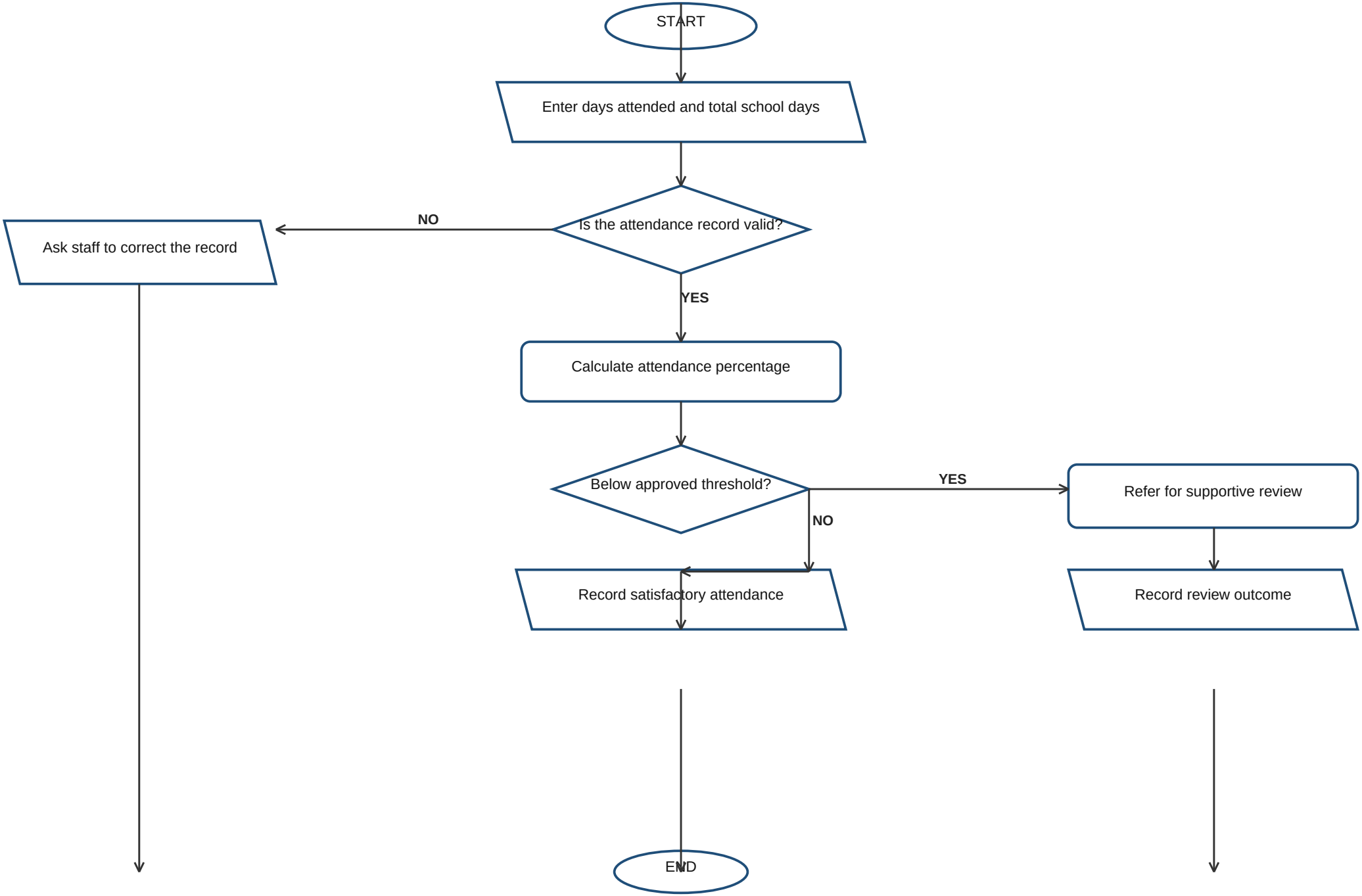
Education Access and Learning Support System (EALSS)



Validate inputs first; compare valid quantities and report the result.

6.2 Flowchart for Attendance Monitoring

Invalid records return for correction; concerns receive supportive review.



The attendance threshold is illustrative and requires approval by the school or authority.

7. Decision Structures, Modules, and Functions

7.1 Decision Structures

Decision structures allow a system to choose between different actions based on conditions.

- Sequence: Instructions are performed in order. For example, the system receives textbook quantities, validates them, and calculates the shortage.
- Selection: The system chooses an action based on a condition. If available textbooks are fewer than required, it reports a shortage; otherwise, it reports sufficient stock.
- Iteration: A process repeats until a condition is met or a collection has been processed. For example, the attendance algorithm can be repeated for every student in a class.

These structures help make the system predictable, organised, and easier to test.

7.2 Modular Design

Modular design means dividing a large system into smaller components, each responsible for a specific task. It improves maintenance, testing, and future development. The proposed system can be divided into the following modules:

Module	Responsibility
Student Records Module	Registration and class details.
Attendance Module	Attendance records and support referrals.
Resource Module	Textbook stock and shortage detection.
Reporting Module	Summaries and reports for authorised users.

7.3 Functions

A function is a reusable operation designed to perform a particular task. In the proposed system, functions could include:

- ValidateAttendance(): checks whether attendance entries are valid.
- CalculateAttendanceRate(): calculates the percentage of school days attended.
- CheckTextbookStock(): compares available textbooks with the required quantity.
- GenerateSchoolReport(): prepares a summary of attendance and learning resources.
- ReferForSupport(): initiates an authorised review when a student needs assistance.

These are logical function names, not implemented source code. Each function would receive the necessary information, perform its assigned task, and return a result or initiate an appropriate action.

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8. Open-Source Software and Digital Public Goods

8.1 Open-Source Software

Open-source software is software whose source code is made available under a licence that allows users to inspect, use, modify, and redistribute it according to the licence conditions. It can help Sierra Leone's education sector reduce dependence on expensive proprietary tools and encourage local technical skills.

- Moodle (https://moodle.org/): an open-source learning management system that supports courses, assignments, and learning materials.
- Koha (https://koha-community.org/): an open-source library management system that can help schools and libraries organise collections.
- LibreOffice (https://www.libreoffice.org/): a free office suite for reports, spreadsheets, and educational documents.
- Linux (https://www.linux.org/): an operating-system family that can run on suitable school computers.

These tools could support different parts of the proposed solution. For example, Moodle could provide learning materials, while Koha could help manage library resources. Open-source software does not mean the entire system has no cost: schools may still need computers, electricity, internet access, technical support, training, and secure hosting.

8.2 Digital Public Goods (DPGs)

Digital Public Goods are digital resources, such as open-source software, open data, open standards, and open content, that meet recognised criteria for openness, privacy, security, and other safeguards. A DPG is more than simply a free digital tool; it should be designed and maintained so that it can be reused and adapted for public benefit.

- Kolibri (https://learningequality.org/kolibri/): an educational technology platform designed to support offline learning.
- Open Educational Resources (https://www.unesco.org/en/open-educational-resources): educational materials that can be accessed and reused under open licences.
- OER Commons (https://oercommons.org/): a collection of learning materials educators can discover and adapt, subject to each resource's licence.

These resources can help address connectivity barriers in Sierra Leone. For example, a school could use an offline learning platform on a local computer network so students can access downloaded educational materials without each device needing an internet connection. Not every open-source product automatically qualifies as a Digital Public Good; formal recognition depends on the relevant criteria and assessment.

8.3 Benefits to Sierra Leone

- Lower software costs: schools can avoid some expensive licence fees.
- Improved access: offline educational platforms can support schools with unreliable connectivity.
- Local adaptation: materials can be adapted to relevant curricula, languages, and community needs where licences permit.

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- Skills development: students and local technology professionals can gain experience with digital systems.
- Collaboration: schools, government institutions, and developers can share improvements and learning resources.
- Sustainability: shared digital resources can be reused across multiple schools rather than developed separately.

Successful implementation would still require training, equipment, technical support, appropriate content, and long-term maintenance.

9. Application of Git and GitHub

Git is a version-control system that records changes to files over time. GitHub is an online platform that hosts Git repositories and supports collaboration, issue tracking, documentation, and project management. Although this assignment does not require source code, Git and GitHub can still manage the system's design documents.

9.1 How Git Can Be Applied

The project team could use Git to track changes to pseudocode documents, Mermaid flowcharts, system requirements, test cases, and technical documentation. For example, the team might create an initial version of the textbook-shortage algorithm, review it, and record an updated version after discovering an input-validation problem. Git preserves the history of changes and helps explain how and why the design evolved.

9.2 How GitHub Can Be Applied

A GitHub repository could contain the following folders:

```
EALSS-Project/
    README.md
    Requirements/
    Pseudocode/
    Flowcharts/
    System-Design/
    Test-Cases/
    Reports/
```

This is an example of a proposed project folder structure, not a computer program. The team could use GitHub issues to record tasks such as designing the attendance flowchart, reviewing pseudocode, or checking accessibility requirements. Pull requests could be used to review proposed changes before they become part of the main project.

9.3 Version-Control Workflow

- Create a GitHub repository for the project.
- Add the system requirements and initial design documents.
- Create a branch for a specific design task.
- Make changes to the relevant document or flowchart.
- Review the changes and request feedback.
- Merge approved changes into the main branch.

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- Maintain a record of completed tasks and outstanding issues.

This process encourages teamwork, accountability, documentation, and continuous improvement.

10. Testing and Evaluation of the Proposed System

Before the system is adopted, its logic should be tested using sample situations. Testing helps determine whether the system produces the expected result for valid and invalid inputs.

Test case	Input or situation	Expected result
1	65 textbooks available; 100 required	Report a shortage of 35.
2	100 textbooks available; 100 required	Report sufficient stock.
3	Negative textbook quantity	Reject the invalid quantity.
4	72 days attended out of 90	Calculate 80% attendance.
5	60 days attended out of 90, with an 80% threshold	Flag for supportive review.
6	Attendance days exceed total school days	Reject the invalid record.

The system should also be tested under realistic school conditions. These tests should establish whether teachers can use it effectively, whether offline materials remain accessible, whether records are protected, and whether reports are accurate.

A pilot project could begin in a small number of schools before expanding to other communities. Evaluation should consider textbook availability, attendance-record accuracy, usage of learning resources, user satisfaction, system reliability, and the cost of maintenance.

11. Contribution to the Sustainable Development Goals

The proposed system supports the United Nations Sustainable Development Goals through improved educational access and better management of school resources.

SDG 4: Quality Education

SDG 4 aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. The system contributes by helping schools identify learning-resource shortages, organise educational materials, and provide access to suitable digital and offline learning content. Attendance monitoring can help schools identify students who may need additional support.

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SDG 9: Industry, Innovation and Infrastructure

SDG 9 focuses on infrastructure, sustainable industrialisation, and innovation. Using affordable digital tools, offline learning systems, and locally maintainable technology can strengthen educational technology infrastructure and encourage local innovation.

SDG 10: Reduced Inequalities

SDG 10 aims to reduce inequality within and among countries. Providing offline learning materials and designing systems for schools with limited internet connectivity can help reduce differences in access to educational resources between well-connected and underserved communities.

The system would contribute to these goals, but achieving them would require broader investment in teachers, school facilities, electricity, connectivity, educational materials, and public policy.

12. Challenges and Proposed Solutions

Challenge	Proposed solution
Unreliable electricity	Use energy-efficient devices, backup power, and solar options where practical.
Limited internet connectivity	Provide offline access and synchronise records when connectivity returns.
Lack of technical skills	Train teachers and school administrators before deployment.
Shortage of computers	Introduce shared computer facilities and prioritise essential functions.
Data privacy risks	Use access controls, backups, secure storage, and minimal data collection.
Maintenance difficulties	Train local support personnel and document the system clearly.
Inaccurate records	Validate inputs and conduct regular checks by authorised staff.
Limited funding	Pilot the system and seek partnerships with government, schools, and development organisations.

These measures should be considered during the planning stage rather than after the system has been deployed.

13. Conclusion

Education challenges in Sierra Leone require practical solutions that consider the needs of students, teachers, school administrators, and communities. This assignment has proposed the Education Access and Learning Support System as a logical approach to improving the management of educational resources and monitoring students' learning needs.

Computational thinking helps analyse the problem through decomposition, pattern recognition, abstraction, and algorithm design. Pseudocode explains the steps of the proposed algorithms, while flowcharts provide a visual representation of their decisions and processes. Decision structures, functions, and modular design make the system

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easier to understand, test, and maintain.

Open-source software and Digital Public Goods can improve access to educational materials while reducing some software costs. Git and GitHub can support version control, documentation, collaboration, and project management, even when the assignment does not require writing source code.

The proposed system contributes to SDG 4, SDG 9, and SDG 10 by supporting quality education, accessible technology, and more equitable access to learning resources. Nevertheless, successful implementation would depend on adequate funding, teacher training, reliable equipment, data protection, and continued technical support.

In conclusion, programming logic is not limited to writing computer programs. It is also a method of analysing problems, organising information, making decisions, and designing practical systems that can contribute to national development.

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14. References

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Submission note: This assignment uses illustrative examples rather than verified national statistics. The Mermaid diagrams are included as diagram definitions and can be rendered or recreated in draw.io. The attendance threshold shown is illustrative and should be approved by the relevant school or authority.

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